

Asymmetric Cross-Reality Collaboration for Urban Sustainability Education

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Figure 1: The Asymmetric Cross-Reality Collaboration setup: The AR user's top-down planning interface, featuring an on-demand menu (a) with a carbon sensor, the VR user's POV feed and policy tokens. (b) The VR user's egocentric perspective for exploring the potential future consequences of policies, with the capability to spawn a VR-to-AR portal for transitioning back to AR planning mode. (c) A city model that includes the location of VR users, connected by a "Ray" to their physical representation. (d) The AR-to-VR portal, activated by selecting a location in the city model, enables a transition to a first-person view of the virtual city.

ABSTRACT

This paper presents a mixed-reality platform that leverages asymmetric cross-reality collaboration between Augmented Reality and Virtual Reality to promote experiential learning and informed decision-making for urban environmental sustainability. Participants take on the role of policymakers addressing urban environmental challenges, engaging in problem-solving, strategy implementation, and reflection. AR users analyze real-time urban conditions, while VR users explore future outcomes of policy interventions, creating a dynamic cause-and-effect feedback loop that stimulates deeper insight into complex decision-making processes. The platform's implementation, including shared point-of-views, persistent avatars, and virtual windows, demonstrates how immersive tools can enable co-located users to collaboratively assess the impact of urban policies, such as public transportation enhancements, on reducing carbon emissions

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and fostering sustainable urban development.

Index Terms: Asymmetric Collaboration, Cross-Reality Interaction, Extended Reality, Virtual Reality, Augmented Reality, Environmental Sustainability

1 INTRODUCTION

Urban areas, hubs of human activity, face significant issues from industrial operations, traffic, and population growth. The urgency of tackling environmental challenges like carbon emissions and resource management highlights the need for impactful educational tools [2]. Given that Virtual and Augmented Reality (VR/AR) technologies offer innovative solutions to promote sustainability and inspire behavioral change [1], we propose a collaborative interface with a learning platform that leverages these technologies to address environmental challenges. Grounded in experiential learning theory, the platform emphasizes active engagement and reflective learning [4]. Participants act as decision-makers, crafting policies to regulate carbon emissions, and observing their long-term impacts. Integrating real-time planning tools with immersive future visualizations, this platform deepens understanding of the consequences of urban environmental policies and promotes pro-environmental engagement.

Our proposed system leverages asymmetric cross-reality perspectives: an AR environment that maps to present-day urban conditions, and a VR environment that projects future scenarios shaped by par-

ticipants' decisions. Asymmetric approaches promote collective problem-solving that exceeds what a single reality could provide [3]. Our design enables seamless switching between current and projected realities and facilitates dynamic feedback loops. Users see the immediate impacts of their policy choices and the long-term consequences of those choices in an immersive virtual future.

Collaboration strategies are flexible, allowing participants to freely choose their approach, using AR and VR simultaneously or switching between them as needed. For example, one participant might focus on modifying the city's transportation networks, such as introducing buses or expanding public transit routes, while another observes the impacts on carbon emissions. Each policy triggers unique outcomes, providing valuable insights into how decision-making shapes environmental changes.

2 ASYMMETRIC PERSPECTIVES FOR COLLABORATIVE SUSTAINABILITY

Since our design allows participants to occupy AR and VR simultaneously and switch freely between these perspectives, it ensures continuous awareness of both present and future states shaped by policy decisions. The AR perspective provides a real-time, top-down view of current conditions, enabling strategic planning, policy implementations and macroscopic analysis of current city conditions. In AR, users retain a critical link to the present-day conditions, facilitating a direct comparison between current and projected states. The VR perspective immerses participants in a first-person, ground-level experience of the city, allowing them to navigate the urban landscape, observe the effects of policy decisions, and witness the city's transformation, providing experiential data regarding the felt impact of decisions, bridging the gap between abstract planning and lived consequences.

Transitions between perspectives occur via virtual portals. When an AR user opens such a portal, the VR environment becomes accessible (Figure 1(d)). Stepping through the portal transports the user into the future, while stepping back transitions them to the present (Figure 1(b)). These portals appear as spatial gateways, deliberately designed to visually anchor the shift between temporal frames and preserve spatial continuity.

3 POLICY CONTROL AND MONITORING CHANGES

An on-demand menu in the AR interface allows participants to enact policy changes that affect the future world-state. Access to this menu is initiated via a pinch-and-swipe-down gesture with the left hand; performing a pinch-and-swipe-up closes it. As illustrated in Figure 1(a), the on-Demand menu provides a mini-map of the city, a summary of current objectives, and a carbon sensor readout that the user immersed in the VR "future" can use to assess changes in carbon emissions in different parts of the city.

To implement interventions in the projected world, the AR user pick up, move, or drop policy tokens (e.g., "electric bus subsidy") onto the city representation. This action triggers corresponding updates in the environment, allowing participants in VR to witness, in real time, how these policies reshape the cityscape, altering textures, traffic density, or sky color to signal shifts in environmental health. The player can see their toolbox with policy tokens, their objective, and a "sensor" that detects the level of carbon in the air. This sensor records the carbon dioxide levels wherever the user currently is. Upon achieving favorable outcomes, the "time-traveling policymaker" in VR observes the sensor data and reports the results. Successful policy enactments immediately reduce or eliminate carbon smog in the future environment. Additional cues reinforce positive environmental transformations achieved through collaborative decision-making. For instance, if the player selects a "public transportation" policy, newly introduced electric buses become visible in VR, emphasizing the tangible impact of the strategy.

4 MANAGING CROSS-REALITY DISSONANCE

A key challenge in cross-reality environments is managing discrepancies in how co-located participants perceive and interact with the shared environment. This phenomenon, known as dissonance, arises when conflicting viewpoints or representations create a lack of coherence between participants' experiences within the same application. For example, while an AR user views a physical room with superimposed digital information, a VR user may inhabit a fully virtual, potentially altered representation of that space, such as an idealized future cityscape. Such mismatched perceptions can lead to confusion and hinder effective collaboration. To address this, we propose two design strategies aimed at reducing perceptual gaps and enhancing seamless interaction:

Persistent Avatar Representations: To address the co-location of AR and VR users, where the AR user can see the VR user physically but not their virtual position, the system tracks the VR user's location within the virtual city and represents it as a capsule in the AR space. A visual ray connects the physical and virtual positions, aligning the VR user's real-world movements with their avatar's actions in the virtual environment (as shown in Figure 1(b)). By grounding the VR user's presence in both the physical and virtual spaces, we ensure that each participant understands the other's perspective and maintains a shared sense of situational awareness.

Virtual Windows: AR participants can open a "Virtual Window" to glimpse the VR user's point of view. Aligning the virtual feed with the AR interface reduces ambiguity about what the VR participant sees and experiences, thereby facilitating clearer communication between participants.

In essence, these strategies work synergistically to ground participants within both the physical and virtual realms, promoting a shared understanding of the environment and each other's actions, ultimately leading to a more cohesive and productive cross-reality experience.

5 DISCUSSION AND CONCLUSION

We have presented a collaborative learning platform leveraging asymmetric cross-reality views and 3D user interfaces to examine urban environmental policies aimed at reducing carbon emissions. The participants learn about Environmental Sustainability through iterative interactions that bridge real-time analysis with future-state observations, allowing them to directly witness the cause-and-effect dynamics of policy decisions. Our approach enables co-located participants to analyze and learn from policy interventions in both present and future projected scenarios. This dual-perspective design promotes richer collaboration and collective problem-solving than either reality could achieve independently. Although environmental challenges persist, our asymmetric viewpoints may significantly enhance understanding of policy outcomes, ultimately contributing to more informed and sustainable urban decision-making.

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